

QUAD CITY, IL, USA

WMO: 725440

Lat: 41.450N

Lon: 90.517W

Elev: 181

StdP: 99.17

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14923

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -20.2	(c) -17.1	(d) -24.7	(e) 0.4	(f) -19.4	(g) -22.0	(h) 0.5	(i) -16.6	(j) 11.8	(k) -5.2	(l) 10.9	(m) -4.4	(n) 3.5	(o) 280	(p) 0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	33.5	24.6	32.1	23.8	30.7	22.9	26.2	31.5	25.2	30.3	24.2	29.0	4.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.6	20.1	29.7	23.6	18.8	28.5	22.7	17.8	27.5	82.2	31.5	77.7	30.3	74.0	29.3	30.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.7	(b) 9.1	(c) 8.2	(e) -25.7	(f) 35.9	(g) 4.8	(h) 1.6	(i) -29.1	(j) 37.0	(k) -31.9	(l) 37.9	(m) -34.5	(n) 38.8	(o) -38.0	(p) 39.9		
(a) 10.7	(b) 9.1	(c) 8.2	(e) -25.9	(f) 28.1	(g) 4.6	(h) 1.1	(i) -29.2	(j) 28.9	(k) -32.0	(l) 29.5	(m) -34.6	(n) 30.1	(o) -37.9	(p) 30.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	-5.0	-2.6	4.2	10.8	16.8	22.2	24.1	23.0	19.1	12.2	4.7	-1.8		
	DBStd	11.40	6.97	6.69	6.37	5.25	4.66	3.58	3.16	2.90	4.37	5.23	5.62	6.31		
	HDD10.0	1656	464	354	203	51	3	0	0	0	1	37	174	369		
	HDD18.3	3356	722	586	441	231	86	9	1	2	43	202	408	625		
	CDD10.0	1912	0	2	21	75	215	364	438	402	273	104	17	2		
	CDD18.3	571	0	0	1	6	39	123	180	146	65	11	0	0		
	CDH23.3	5438	0	0	11	83	384	1176	1824	1248	610	101	1	0		
	CDH26.7	1922	0	0	0	16	104	432	723	418	204	23	0	0		
(14) Wind	WSAvg	3.6	4.2	4.1	4.3	4.6	3.8	3.2	2.8	2.6	2.9	3.5	3.9	3.9		
(15) Precipitation	PrecAvg	975	40	38	70	98	111	118	118	106	90	74	61	52		
	PrecMax	1432	112	90	189	287	290	336	299	387	360	217	172	127		
	PrecMin	451	8	5	16	24	8	26	10	9	0	0	7	0		
	PrecStd	191	24	23	41	47	59	70	67	73	67	48	39	30		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.6	17.4	25.0	28.8	32.1	34.4	36.2	34.2	33.2	29.7	22.3	16.4		
		MCWB	9.3	11.8	17.1	17.4	21.8	24.6	26.2	24.5	23.3	21.3	14.8	14.7		
	2%	DB	9.6	13.0	20.7	25.9	29.4	32.7	33.8	32.3	31.2	26.1	18.7	12.8		
		MCWB	6.8	9.2	14.4	16.7	20.4	23.5	25.3	24.6	22.4	18.2	13.3	10.3		
	5%	DB	6.2	9.4	17.2	22.9	27.4	31.0	32.2	30.6	29.1	23.5	16.1	9.4		
		MCWB	4.1	6.5	12.7	14.9	19.2	22.6	24.5	23.5	21.0	17.0	11.9	6.8		
	10%	DB	3.8	6.3	14.0	20.3	25.4	29.3	30.5	29.0	27.0	20.9	13.7	6.5		
		MCWB	2.1	4.1	9.6	13.4	18.3	21.7	23.5	22.4	20.0	15.7	10.0	4.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	13.4	18.1	19.6	23.8	26.2	27.9	26.9	25.0	22.1	16.7	14.9		
		MCDB	12.3	15.4	23.5	26.1	30.0	32.2	33.6	32.1	30.8	28.0	20.4	16.6		
	2%	WB	6.7	9.5	15.5	17.9	21.9	25.0	26.7	25.6	23.7	20.1	14.4	10.4		
		MCDB	9.2	12.7	19.4	23.4	27.5	30.7	32.0	30.5	29.1	24.0	17.3	12.2		
	5%	WB	4.2	6.5	12.9	16.2	20.6	23.9	25.6	24.6	22.5	18.1	12.5	7.2		
		MCDB	6.2	9.1	17.1	21.8	25.5	29.2	30.7	28.9	27.1	21.9	15.7	9.1		
	10%	WB	2.2	4.1	9.9	14.3	19.3	22.8	24.6	23.7	21.3	16.4	10.1	4.4		
		MCDB	3.7	6.1	13.9	19.5	23.9	27.6	29.3	27.6	25.5	20.5	13.2	6.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	9.0	10.4	11.7	11.5	11.1	10.8	10.6	12.2	11.2	9.6	8.1		
		MCDBR	11.4	13.0	14.3	15.9	13.5	12.8	11.8	11.8	13.8	14.2	13.3	11.4		
	5% WB	MCWBR	9.2	9.6	9.1	8.9	6.8	5.8	5.4	5.7	6.4	7.8	9.1	9.0		
		MCWBR	10.8	12.0	12.8	13.3	11.9	11.2	10.5	10.4	11.4	11.5	11.8	10.5		
(40) Clear-Sky Solar Irradiance	taub		0.292	0.305	0.343	0.390	0.424	0.427	0.445	0.426	0.396	0.348	0.304	0.292		
	taud		2.416	2.391	2.377	2.252	2.222	2.266	2.197	2.252	2.307	2.443	2.500	2.436		
	Ebn at Noon		863	906	905	879	855	851	830	835	836	845	845	831		
	Edh at Noon		79	94	108	132	139	134	142	130	115	90	73	72		
(44) All-Sky Solar Radiation	RadAvg		1.76	2.75	3.74	4.67	5.47	6.15	6.10	5.39	4.58	3.02	2.00	1.47		
	RadStd		0.21	0.28	0.30	0.41	0.40	0.54	0.40	0.35	0.38	0.35	0.17	0.16		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page